

Module 7

Financial Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

$$\begin{aligned}E(R_A) &= R_f + [E(R_m) - R_f] \times \beta_A \\0.10 &= 0.03 + (0.12 - 0.03) \times \beta_A \\ \beta_A &= 0.07 / 0.09 = 0.78\end{aligned}$$

$$\begin{aligned}E(R_B) &= R_f + [E(R_m) - R_f] \times \beta_B \\0.13 &= 0.03 + (0.12 - 0.03) \times \beta_B \\ \beta_B &= 0.10 / 0.09 = 1.11\end{aligned}$$

$$\begin{aligned}E(R_C) &= R_f + [E(R_m) - R_f] \times \beta_C \\0.16 &= 0.03 + (0.12 - 0.03) \times \beta_C \\ \beta_C &= 0.13 / 0.09 = 1.44\end{aligned}$$

Answer 1(b)

Coefficient of variation ("CV") is calculated as standard deviation divided by the expected return.

$$CV_A = 2.0\% / 10\% = 0.20$$

$$CV_B = 2.2\% / 13\% = 0.17$$

$$CV_C = 3.0\% / 16\% = 0.19$$

Answer 1(c)

Stock B should be selected as it has the lowest risk per unit of expected return (i.e. $CV_B = 0.17$).

Answer 1(d)(i)

The expected return of the portfolio is:
 $(1 / 3) \times 0.1 + (1 / 3) \times 0.13 + (1 / 3) \times 0.16 = 0.13 = 13.0\%$

Answer 1(d)(ii)

The total risk of the portfolio can be measured by the standard deviation of the portfolio.

However, the standard deviation of the portfolio is generally less than the weighted average of an individual stock's standard deviation. They will be the same only when all the stocks in the portfolio are perfectly positively correlated (i.e., their returns move in the same direction and by the same degree).

As Alder picked the three stocks from different industries, their returns will move in different directions and by different degrees (i.e., they are uncorrelated with each other). As such, the total risk of the portfolio cannot be measured by the weighted average of an individual stock's standard deviation.

Answer 1(e)

Systematic risk, sometimes called market risk, is the risk of being in the market rather than in any specific investment. It affects a large number of companies. In contrast, unsystematic risk only affects a particular company or a small number of companies.

Systematic risk, also called non-diversifiable risk, is the risk that cannot be mitigated through diversification while the unsystematic risk may be eliminated through the use of a diversified portfolio with stocks from different industries.

Asset market only awards systematic risk but not unsystematic risk, and this is why finance is particularly concerned about systematic risk when evaluating the relationship between risk and return.

An unexpected movement in interest rates is an example of systematic risk. Strike activity in a factory is an example of unsystematic risk.

Note: Any other valid examples are also acceptable.

Answer 2(a)

Pro forma statement of cash flows for the year ended 31 December 2021

	HK\$'000
Income before interest and tax	335
Depreciation	85
Interest	(45)
Tax (42 + 49 - 80)	(11)
Dividends (173 + 96 - 205)	(64)
Non-current asset purchases (1,180 - 1,155 + 85)	(110)
Increase in inventories (152 - 120)	(32)
Decrease in accounts receivable (180 - 165)	15
Increase in accounts payable (188 - 150)	<u>38</u>
Projected increase in cash over the year	211
Cash as at 31 December 2020	<u>220</u>
Cash as at 31 December 2021 (Forecast)	<u>431</u>

Answer 2(b)

DuPont equation

Return on equity ("ROE") = Net profit margin x Total asset turnover x Equity multiplier

ROE = (Net income / Net sales) x (Net sales / Total Assets) x (Total assets / Total equity)

Net profit margin = HK\$241,000 / HK\$2,050,000 = 11.76%

Total asset turnover = HK\$2,050,000 / HK\$1,928,000 = 1.06

Equity multiplier = HK\$1,928,000 / HK\$955,000 = 2.02

ROE = 11.76% x 1.06 x 2.02 = 25.18%

Answer 2(c)

The DuPont equation uses financial ratios to determine the financial health of a company by taking three steps. Management will first use DuPont ratios to assess the financial health of a company. Management will then correct any identified problems. Finally, management will monitor the financial performance continually over time.

The DuPont equation is broken down into three separate components:

ROE = Net profit margin × total asset turnover × Equity multiplier

It identifies three areas to maximise ROE: (1) profit on sales, (2) how efficiently company assets are used and (3) use of financial leverage. A single ratio monitors each of these areas and together these ratios make up the DuPont equation.

For example, if a company's ROE increase is due to an increase in the net profit margin or total asset turnover, this reflects that the company is very healthy. However, if the ROE increase is caused by an increase in equity multiplier and the company is not under-leveraged, this could be a negative sign for the company.

Answer 3(a)

$K_E = (\text{HK\$}2.5 \times 1.06) / \text{HK\$}30 + 6\% = 14.83\%$

Answer 3(b)

$W_E = E/V = \text{HK\$}2,000,000 / \text{HK\$}10,000,000 = 20\%$

$W_P = P/V = \text{HK\$}3,000,000 / \text{HK\$}10,000,000 = 30\%$

$W_D = D/V = \text{HK\$}5,000,000 / \text{HK\$}10,000,000 = 50\%$

$WACC = 14.83\% \times 20\% + 8\% \times 30\% + 6\% \times (1 - 16.5\%) \times 50\% = 7.87\%$

Answer 3(c)

$PV = \text{HK\$}1.2 \text{ million} / 0.0787 = \text{HK\$}15.25 \text{ million}$

$NPV = - \text{HK\$}10 \text{ million} + \text{HK\$}15.25 \text{ million} = + \text{HK\$}5.25 \text{ million}$

Since the net present value is positive, the project should be accepted.

Answer 3(d)

Advantages of debt financing compared to equity financing:

Debt financing will not dilute the control of the existing shareholders.

Companies only need to repay the interest and principal of the debt due to lenders and lenders will have no claim on the profits of the company.

Interest is often tax deductible, which lowers the cost of the debt.

Debt financing is cheaper and easier to obtain than equity financing.

Disadvantages of debt financing compared to equity financing:

Interest costs are fixed and must be paid even though the company is not making any profits.

Debt must be repaid on maturity but equity financing does not.

There may be debt covenants to restrict the activities of the company.

Lenders may require the company to put up assets as collateral.

The higher the debt amount, the riskier the company may be considered by investors.

Answer 3(e)

The pecking order theory is different from other theories that try to find an optimal capital structure. This theory recognises that different types of capital have different costs which determine a pecking order in financing choices.

Managers have a choice when deciding how to finance a new project. Managers first choose the least expensive capital then move on to capital that costs more, and so on, in a simple and efficient manner.

Based on the established pecking order, the cheapest funds are all retained earnings generated internally if they are available, rather than going out to spend time to persuade creditors or equity investors for providing extra capital.

Then debt is the next cheapest funds to obtain but is still relatively inexpensive. It also provides tax benefits on interest payments.

Raising capital by issuing equity can be expensive and should be used as a last resort. The time and associated costs of issuing equity are much higher than the comparable costs for debts. Moreover, the filings required by regulatory agencies on issuing equity are more complicated and time consuming.

Stock markets often react badly to announcements that companies are issuing additional equity, because these equity issues can be seen as evidence that the company is not profitable enough to fund investments and capital expenditure internally.

In addition, issuing equity is interpreted as a sign that the management of the company believes that the shares of the company are overvalued. In turn, the share price will fall and makes future equity issues more expensive.

Answer 4(a)(i)

Incremental cost of purchasing component X = HK\$(200 – 70 – 50 – 24) = HK\$56

Incremental cost per machine hour = HK\$56 / 5 = HK\$11.2

Ranking order of cheapest = 3

Incremental cost of purchasing component Y = HK\$(140 – 60 – 40 – 15) = HK\$25

Incremental cost per machine hour = HK\$25 / 4 = HK\$6.25

Ranking order of cheapest = 1

Incremental cost of purchasing component Z = HK\$(78 – 30 – 28 – 11) = HK\$9

Incremental cost per machine hour = HK\$9 / 1 = HK\$9

Ranking order of cheapest = 2

In order to make all components in-house, it would take $(3,000 \times 5) + (1,000 \times 4) + (2,000 \times 1) = 21,000$ hours.

As there are only 15,800 hours available, there is a shortage of 5,200 hours $(21,000 - 15,800)$. The shortage of 5,200 hours should be allocated in the ranking order:

Machine hour allocated to component Y: $1,000 \times 4 = 4,000$ hours which is less than 5,200 hours.

Number of units of component Y to be purchased = 1,000 units

Remaining hours after allocation to component Y = $5,200 - 4,000 = 1,200$ hours which will be allocated to component Z.

Number of units of component Z to be purchased = $1,200 / 1 = 1,200$ units

Therefore, 1,000 units of component Y and 1,200 units of component Z should be purchased from the external supplier.

Answer 4(a)(ii)

The quality of the components produced by the supplier.

The supplier's ability to increase its production facility's capacity to meet DML's order.

Whether the supplier will accept the local currency of DML for the purchase price of the components.

The lead time between placing the order and actual delivery.

Answer 4(b)(i)

After three months, the spot rate would be $\text{HK\$}8.80 \times (1 + 5\%) = \text{HK\$}9.24$.

According to the forward contract, DML has to sell EUR200,000 using the forward rate of HK\$8.60. If DML has not made the sales and will not receive the EUR200,000 in three months, it will have to buy EUR200,000 from the market to fulfil the forward contract. As such, the loss on the forward contract will be $\text{HK\$}(9.24 - 8.60) \times 200,000 = \text{HK\$}128,000$.

Answer 4(b)(ii)

At the time DML enters into the forward contract, it has no idea what the exchange rate will be in the future. By making the overseas sales and owning the EUR200,000 receivable, DML will receive more in HK\$ if EUR appreciates and will receive less in HK\$ if EUR depreciates in three months.

If EUR appreciates by 5% in three months, DML will receive more in HK\$ (i.e., $200,000 \times \text{HK\$}9.24 = \text{HK\$}1,848,000$) on the receivable. However, the total cash receipts in HK\$ will be offset by the loss on the forward contract (i.e., HK\$128,000 as calculated in Question 4(b)(i) above). The total net amount received by DML will be HK\$1,720,000 ($\text{HK\$}1,848,000 - \text{HK\$}128,000$).

However, if EUR depreciates to a rate lower than the forward rate in three months, DML will receive less in HK\$ on the receivable. Similarly, the total cash receipts in HK\$ will be compensated by the gain on the forward contract. The total net amount received by DML will still be HK\$1,720,000.

The purpose of hedging with a forward contract is to ensure the value of the EUR200,000 receivable will be fixed in HK\$ (i.e., HK\$1,720,000 by converting the EUR200,000 at the contracted rate of HK\$8.60) no matter whether the exchange rate between EUR and HK\$ will go up or down in three months.

As such, the future exchange rate is locked at the forward contract rate of HK\$8.60 per EUR and the currency risk can be mitigated.

Answer 4(b)(iii)

Another major risk is the credit risk that DML's customer may not pay in three months.

There are several ways to mitigate credit risk. Performing a credit check for the customer is the first step.

DML can reduce its credit risk by obtaining credit insurance on the invoice issued to the customer.

Shortening the payment may be another alternative.

DML may also consider requesting a personal guarantee or a third-party guarantee from the customer.

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END OF EXAMINATION PAPER

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